

This assignment will not be graded and is only for practice.

Level 1:

1) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ be a matrix (where the first row denotes the top most row, and the

1 point

ordering of the rows is in the order top to bottom). Among the given set of options, identify the correct statements.

- ☐ The first non-zero element in the first row is 3.
- ☐ The first non-zero element in the second row is 1.
- ☐ There is a non-zero element in the third row.
- ☐ Since there is a row with all elements as zero, $\det(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The first non-zero element in the second row is 1.

Since there is a row with all elements as zero, $\det(A) = 0$.

Let $I_{3 \times 3}$ denote the identity matrix of order 3. Answer questions 2 and 3 about the set S defined as :

$$S = \left\{ A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \right.$$

$$E = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, G = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, J = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, K = I_{3 \times 3}, L = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \}.$$

2) If S_1 is the subset of S consisting of all the matrices in S that are in row echelon form, then **1 point**
choose the correct option from the following.

- ☐ $S_1 = S$
- ☐ $S_1 = \{A, B, C, D, F, G, I\}$
- ☐ $S_1 = \{B, C, D, E, F, G, H, I, J, K, L\}$
- ☐ $S_1 = \{C, D, E, F, G, H, I, J, K\}$
- ☐ $S_1 = \{C, D, E, F, G, H, J, K, L\}$
- ☐ $S_1 = \{C, D, F, G, H, J, K, L\}$
- ☐ Cardinality of S_1 is 7.
- ☐ Cardinality of S_1 is 8.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S_1 = \{C, D, F, G, H, J, K, L\}$

Cardinality of S_1 is 8.

3) If S_2 is the subset of S consisting of all the matrices in S that are in reduced row echelon form, then choose the correct option from the following. **1 point**

- ☐ $S_2 = \{A, C, D, F, G, L\}$
- ☐ $S_2 = \{B, C, D, E, F, G, H, J, K\}$
- ☐ $S_2 = \{C, D, G, H, J, K, L\}$
- ☐ $S_2 = \{C, D, H, J, K, L\}$
- ☐ $S_2 = \{C, D, H, I, J, K, L\}$
- ☐ Cardinality of S_2 is 7.
- ☐ Cardinality of S_2 is 6.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$S_2 = \{C, D, H, J, K, L\}$$

Cardinality of S_2 is 6.

4) Consider the following system of linear equations:

1 point

$$0x_1 + x_2 + 0x_3 + 0x_4 = 1$$

$$0x_1 + 0x_2 + 0x_3 + 0x_4 = 1$$

$$x_1 + x_2 + 0x_3 + 0x_4 = 1$$

$$0x_1 + 0x_2 + x_3 + x_4 = 1.$$

Choose the set of correct options.

- ☐ The system of linear equations has a solution
- ☐ The system of linear equations has no solution.
- ☐ $\det(A) = 0$, where A is the coefficient matrix of the given system of linear equations.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has no solution.

$\det(A) = 0$, where A is the coefficient matrix of the given system of linear equations.

5) Consider a system of linear equations:

1 point

$$0x_1 + x_2 + 0x_3 + 0x_4 = 1$$

$$0x_1 + 0x_2 + x_3 + 0x_4 = 1$$

Choose the set of correct options.

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- ☐ x_1 and x_2 are dependent variables.
- ☐ x_2 is a dependent variable.
- ☐ x_3 and x_4 are independent variables.
- ☐ x_4 is an independent variable.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_2 is a dependent variable.

x_4 is an independent variable.

Level 2:

6) Suppose a system of linear equations consists of only one equation and four variables as follows:

$$x_1 + x_2 + x_3 + x_4 = a$$

where a is a constant. Find out the number of independent variables.

[Hint: "Think about" how many variables can be expressed in terms of the other variables in the given system of linear equations?]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Let $[A|b]$ denote the augmented matrix of the system of linear equations

1 point

$$2x_1 + x_2 = 3$$

$$x_1 + 3x_2 = 4$$

where, $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Let the matrix

$$\left[\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b \end{array} \right]$$

denote the reduced row echelon form of the augmented matrix corresponding to the system of linear equations above. Which of the following option(s) is (are) correct?

- ☐ The values of a and b cannot be determined from the given information.
- ☐ $a = b$ but their exact values cannot be determined from the given information.
- ☐ $a = b = 1$
- ☐ $a = 2$ and $b = 3$
- ☐ The solutions for x_1 and x_2 are not unique.
- ☐ $x_1 = x_2 = 1$, and the system of linear equations has a unique solution.
- ☐ $x_1 = x_2 = 1$ is the solution. However, it is not possible to determine whether the system of equations has a unique solution or not from the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$a = b = 1$$

$x_1 = x_2 = 1$, and the system of linear equations has a unique solution.

8)

1 point

Let $Ax = b$ be a matrix representation of a system of linear equations, where A is a 4×4 matrix, $x =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ 0 \\ 0 \end{bmatrix}.$$

Let A be $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$

Which of the following options are correct?

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- ☐ x_1 is dependent on x_2 .
- ☐ x_3 is dependent on x_4 .
- ☐ x_2 is an independent variable.
- ☐ The solution of the system of linear equations (if it exists) is unique.
- ☐ There exist infinitely many solutions for the given system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_3 is dependent on x_4 .

x_2 is an independent variable.

There exist infinitely many solutions for the given system of linear equations.

Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ be a matrix (where the first row denotes the top most row, and the ordering of the rows

is from top to bottom). Consider the system of linear equations given by $Ax = b$ where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b =$

4
3
0

. Answer the following questions.

9) Find the number of independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) If $x_1 = 0$ is given, then find out the number of solutions of the given system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/10